

Practical – 4

Practical 4 AIM:

Find the Maximum Temperature Recorded for Each Year of NCDC Data using MapReduce

Problem Definition :

A meteorological research organization maintains historical weather records collected from various weather stations worldwide. The organization uses Hadoop to analyze large climate datasets and study long-term temperature trends. As a Data Analytics Engineer, you are required to process the National Climatic Data Center (NCDC) dataset and identify the maximum temperature recorded for each year using MapReduce.

Task 1: Upload the NCDC weather dataset to HDFS and examine the structure of the weather records.

Task 2: Develop and implement the Mapper program to extract the year and temperature values from each record and generate intermediate key-value pairs.

Task 3: Develop and implement the Reducer program to determine the maximum temperature recorded for each year.

Task 4: Execute the MapReduce job on Hadoop and store the results in HDFS. Verify the correctness of the output generated.

Task 5: Analyze the yearly maximum temperature trends and identify the hottest year in the dataset. Discuss the advantages of using MapReduce for large-scale climate data analysis.

Output:

```

bhumi@kali: ~/practical4
Session Actions Edit View Help
(bhumi@kali)-[~/practical4]
$ hdfs dfs -put -f ncdc.txt /practical4/input/
(bhumi@kali)-[~/practical4]
$ hdfs dfs -ls /practical4/input
Found 1 items
-rw-r--r--  1 bhumi supergroup      120 2026-09-12 13:50 /practical4/input/ncdc.tx
t
(bhumi@kali)-[~/practical4]
$ hdfs dfs -cat /practical4/input/ncdc.txt
1901,32
1901,45
1901,38
1902,40
1902,48
1902,42
1903,35
1903,50
1903,44
1904,41
1904,55
1904,47
1905,39
1905,52
1905,46

```

```

bhumi@kali: ~/practical4
Session Actions Edit View Help
2018-11-11 11:52 Reduce output records=5
2018-11-11 11:52 Spilled Records=30
2018-11-11 11:52 Shuffled Maps =1
2018-11-11 11:52 Failed Shuffles=0
2018-11-11 11:52 Merged Map outputs=1
2018-11-11 11:52 GC time elapsed (ms)=61
2018-11-11 11:52 Total committed heap usage (bytes)=274866176
2018-11-11 11:52 Shuffle Errors
2018-11-11 11:52 BAD_ID=0
2018-11-11 11:52 CONNECTION=0
2018-11-11 11:52 IO_ERROR=0
2018-11-11 11:52 WRONG_LENGTH=0
2018-11-11 11:52 WRONG_MAP=0
2018-11-11 11:52 WRONG_REDUCE=0
2018-11-11 11:52 File Input Format Counters
2018-11-11 11:52 Bytes Read=120
2018-11-11 11:52 File Output Format Counters
2018-11-11 11:52 Bytes Written=40
(bhumi@kali)-[~/practical4]
$ hdfs dfs -cat /practical4/output/part-r-00000
1901 45
1902 48
1903 50
1904 55
1905 52
(bhumi@kali)-[~/practical4]
$

```

Key Questions / Analysis

1. How is weather data parsed and processed using MapReduce?

The Mapper reads each weather record as a line of text and splits it into different fields. In our sample dataset, the year and temperature are separated by a comma. The Mapper extracts the **year as the key** and **temperature as the value**. For example, 1904,55 becomes:

(1904, 55)

These key-value pairs are then passed to the Reducer.

2. What key-value pairs are generated by the Mapper?

The Mapper generates the **year as the key** and the **temperature as the value**. For example, if the input contains:

1901,32
1901,45
1902,40
1902,48

The Mapper produces:

(1901, 32)
(1901, 45)
(1902, 40)
(1902, 48)

All temperature values belonging to the same year are grouped together before being sent to the Reducer.

3. How does the reducer identify the maximum temperature for a given year?

The Reducer receives one year and all its corresponding temperature values. It compares these values one by one and keeps the **largest temperature**. Finally, it outputs the year along with its maximum temperature.

4. What challenges arise when processing large climate datasets?

Some challenges include **very large data size, missing or incorrect values, different data formats, and processing time**. The dataset may contain millions of weather records, so efficient parsing and processing are important. Network communication between Map and Reduce tasks can also affect performance. MapReduce helps handle these challenges by dividing the work among multiple machines.

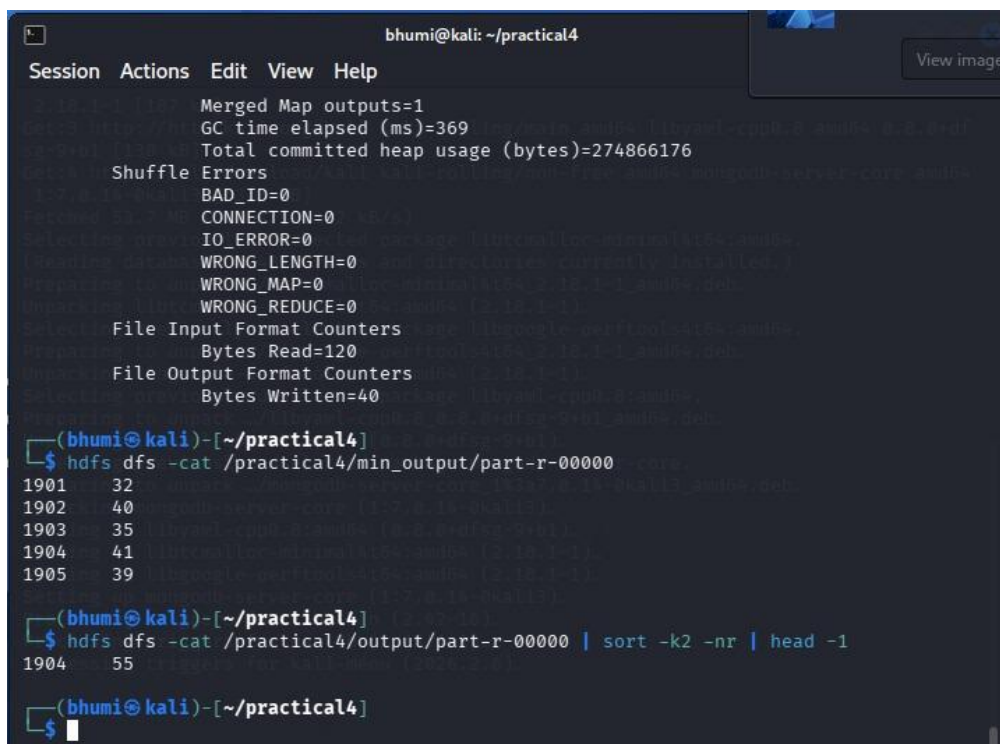
5. How does Hadoop improve scalability and performance for weather data analytics?

Hadoop scales by **dividing the large dataset into smaller parts** and processing them in parallel across multiple nodes. Multiple Mapper tasks can process different portions of the weather data simultaneously. Reducers then combine and analyze the intermediate results to generate the final yearly maximum temperatures. Therefore, MapReduce is suitable for large climate datasets because adding more machines can increase the processing capacity.

Supplementary Problems

Find the minimum temperature recorded for each year. Calculate average yearly temperatures. Identify the year with the highest recorded temperature overall.

Output:



```

bhumi@kali: ~/practical4
Session Actions Edit View Help
Merged Map outputs=1
GC time elapsed (ms)=369
Total committed heap usage (bytes)=274866176
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=120
File Output Format Counters
Bytes Written=40

(bhumi@kali)~/practical4
$ hdfs dfs -cat /practical4/min_output/part-r-00000
1901 32
1902 40
1903 35
1904 41
1905 39

(bhumi@kali)~/practical4
$ hdfs dfs -cat /practical4/output/part-r-00000 | sort -k2 -nr | head -1
1904 55

(bhumi@kali)~/practical4
$
  
```

Key Skills to be Addressed

- MapReduce Programming
- Data Parsing
- Key-Value Pair Processing
- Data Aggregation
- Maximum Value Calculation
- HDFS Data Handling
- Large-Scale Data Processing
- Data Analysis

Tools /Technology

Hadoop 3.x | HDFS | Java JDK 8/11 | Eclipse/IntelliJ IDEA | Linux/Ubuntu Environment | NCDC Weather Dataset | MapReduce